

## EXPERIENCE

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- Duferco** Lugano, Switzerland  
*Software Engineer* 03/2025 - Present
  - Defined the **company-wide backend standards** (auth, observability, data modeling) and shipped them as a shared **Python** platform library and reusable **CI/CD** pipelines adopted across services.
  - Deliver and maintain **8+ production FastAPI/Python services** to the same engineering bar — layered architecture, typed **Pydantic** contracts, full observability, and comprehensive **test suites** (hundreds of tests per service).
  - Cut a core **scheduling/optimization batch** from **30 to 2 minutes (15x)** by reworking the hot path, cutting the heaviest query latency **96%**, and parallelizing I/O.
  - Designed a middleware bridging cloud services to a legacy **AS/400** (IBM DB2 for i) ERP over HTTP — **Apache Arrow** streaming and atomic batch transactions — removing client-side DB drivers across services.
  - Built and maintained the team's **AI-agent development platform** — a private plugin/skill **marketplace** (Claude Code and Codex) plus agentic workflows that auto-open **PRs** to keep a cross-service knowledge base current — standardizing AI-driven development across projects.
  - Built an **LLM** document-extraction service (**Claude** on AWS Bedrock) converting business documents into validated structured data.
- Jobtome** Stabio, Switzerland  
*Software Engineer* 11/2021 - 02/2025
  - Built **search APIs** in **Python** over **Elasticsearch**, sustaining **1,000 requests/second**.
  - Architected a **Python** job-crawler platform on **GCP** — **5,000+ crawlers, 300k+ jobs/day** — plus **Apache Beam** pipelines ingesting **millions of records/day** (Pub/Sub to BigQuery).
  - Built and deployed an **NLP** classifier (**BERT, CNN**) on **Vertex AI**, processing **about 1M jobs/day** at **95% accuracy**.
- Polytechnic University of Milan** Milan, Italy  
*Research Fellow* 10/2019 - 10/2021
  - Built **NLP** data pipelines analyzing **3M+ Italian social-media conversations** on vaccine discourse, and insights from reviews across **100 museums**.

## SKILLS

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- Languages:** Python, Java, C++, SQL
- Backend:** FastAPI, Pydantic, asyncpg, PostgreSQL, REST, OAuth2/JWT
- Data & ML:** Pandas, NumPy, PyTorch, BERT/Transformers, LLMs (Claude, AWS Bedrock), Apache Beam, BigQuery
- Cloud & Infra:** AWS, GCP, Docker, Kubernetes, OpenTelemetry, CI/CD, distributed systems

## EDUCATION

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- Polytechnic University of Milan** Milan, Italy  
*M.Sc. in Computer Science and Engineering; Final grade: 110 with honors/110* 09/2016 – 04/2019
- Polytechnic University of Milan** Milan, Italy  
*Bachelor Engineering of Computing Systems* 09/2013 – 09/2016

## SELECTED PUBLICATIONS

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- Pavanetto, Brambilla:** *Generating Realistic Web-Navigation Paths with RNNs and GANs*. ICWE 2020.
- Di Giovanni, Pavanetto, et al.:** *Content-based Classification of Vaccine-related Stances on Twitter (Italian)*. ICWSM 2021.